



4.3.4. Fourier Filtering - Practice

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4.3.4. Fourier Filtering - Practice Assessment

Checkboxes

0 points possible (ungraded)

Fourier filtering involves taking the Fourier Transform of an image and performing operations in the Fourier domain before switching back to the spatial domain.

Which of the following statements are true?

Check all that apply

☒ the Fourier transform is reversible ✓☐ the Fourier domain is also called the spatial domain☒ the Fourier transform of an image is axially symmetric around the center ✓☒ the farther we are from the center of the Fourier image, the higher the frequency in the image ✓

i Answers are displayed within the problem

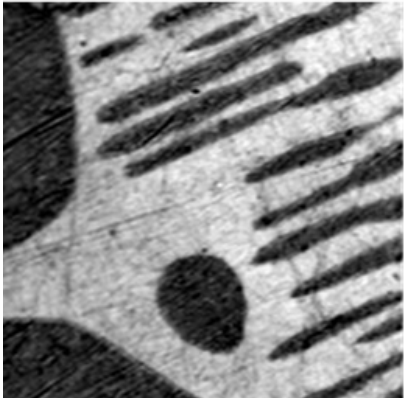
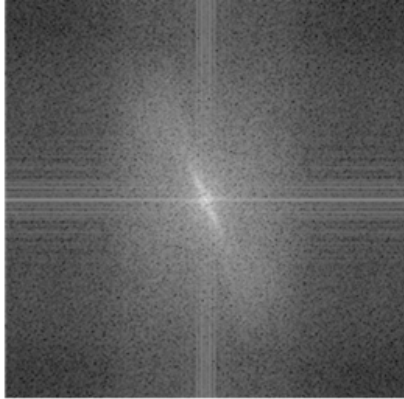
Fourier Filters

0 points possible (ungraded)

Keyboard Help

PROBLEM

Suppose we have the FFT (top) of the image shown here (bottom)



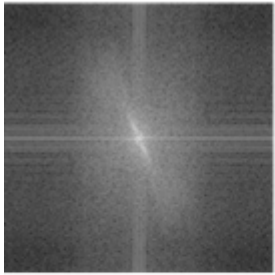
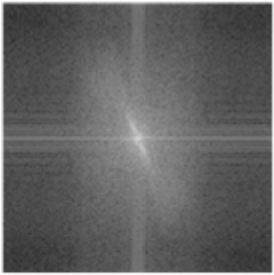
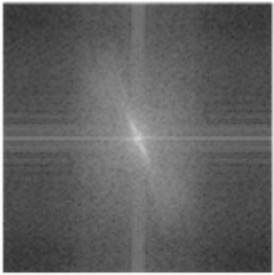
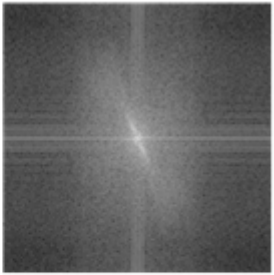
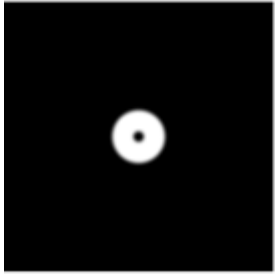
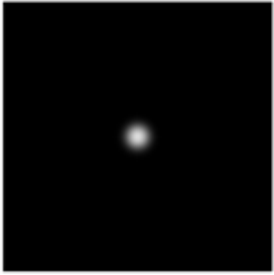
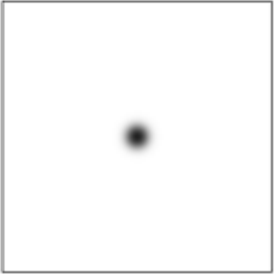
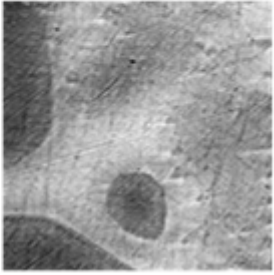
We then choose to apply fourier filtering in order to enhance or suppress features based on their angle or frequency. **Drag and drop the effect of the filter onto the right corresponding images.**

a bandpass filter
that tries to
enhance average
sized objects,
while removing
noise and large
objects

a stripe filter that
removes average
sized objects at an
angle

a lowpass filter,
that aims to
enhance larger
objects

a highpass filter
that aims to
enhance the
smallest features
in the image

			
X	X	X	X
			
↓FFT ⁻¹	↓FFT ⁻¹	↓FFT ⁻¹	↓FFT ⁻¹
			


Reset

FEEDBACK

 Great! Fourier filtering is not necessarily intuitive, but consitutes a great way to apply complex filtering to images based on their content.

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